Quantum Mechanics II — Exercise Sheet 6

Spectra, resolvents and bands

Variant: Questions only

Exercise 1 (Dense eigenvalues). Let $(q_n)_{n\geq 1}$ enumerate the rationals in $[0,1]$ without repetition. On $\ell^2(\mathbb{N})$ let $(Ac)_n = q_n c_n$. Determine the spectrum and the eigenvalues. For an irrational $\lambda \in (0,1)$ construct a Weyl sequence from coordinate vectors and explain why λ is not an eigenvalue.

Exercise 2 (The resolvent norm of a two-level operator). On $\mathbb{C}^2$ let $A = \operatorname{diag}(-2,3)$. For $z \in \mathbb{C}\setminus\{-2,3\}$ compute $(A-z)^{-1}$ and its norm. At $z = \mathrm{i}$ compare the exact norm with $1/|\operatorname{Im} z|$. State why this finite example does not prove equality in the lecture's resolvent bound for every self-adjoint operator.

Exercise 3 (A dimerized chain). A two-site periodic chain has Bloch matrix

$$h(k) = \begin{pmatrix} 0 & \kappa_1 + \kappa_2 e^{-\mathrm{i}k} \\ \kappa_1 + \kappa_2 e^{\mathrm{i}k} & 0 \end{pmatrix}, \qquad -\pi \leq k \leq \pi,$$

where $\kappa_1, \kappa_2 > 0$. The Fourier decomposition identifies the chain Hamiltonian with multiplication by this matrix; take this identification as given. (i) Find the two bands and their gap. (ii) Treat $\kappa_1 = \kappa_2$ separately.

Exercise 4 (A resolvent calculation with a source). On $L^2(\mathbb{R})$ let A be multiplication by x^2 on its maximal domain. For $f = \mathbf{1}_{[-1,1]}$ solve $(A+1)u = f$. Verify directly that $u \in D(A)$ and compute $\|u\|^2$. Use $\int \mathrm{d}x/(1+x^2)^2 = x/[2(1+x^2)] + \arctan x/2$.